Module 2 — Foundations in Chemistry

2.1 Atoms, Ions and Compounds

Atomic Structure

Atom: The smallest particle of an element that can take part in a chemical reaction, consisting of a nucleus (protons and neutrons) surrounded by electrons in energy levels. The nucleus is extremely small (radius ~10^{-15} m) compared to the atom as a whole (~10^{-10} m), yet contains virtually all the mass.

Particle	Relative Mass	Relative Charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	1/1836 (negligible)	−1	Shells/orbitals around nucleus

Atomic number (Z) = number of protons. This defines which element an atom is and determines its position in the periodic table. **Mass number (A)** = number of protons + neutrons. In a neutral atom, the number of electrons equals the number of protons. Ions form when this balance is disrupted.

Isotopes: Atoms of the same element (same number of protons) with different numbers of neutrons, and therefore different mass numbers. Isotopes have identical chemical properties (because they have the same electron configuration and nuclear charge) but different physical properties (e.g. density, rate of diffusion, radioactive behaviour).

Relative Atomic Mass and Mass Spectrometry

Relative atomic mass (A_r): The weighted mean mass of an atom of an element compared to 1/12 of the mass of a carbon-12 atom. It is a dimensionless quantity (no units) because it is a ratio.

$$A_r = \Sigma(\text{isotopic mass} \times \text{percentage abundance}) / 100$$

Relative atomic masses are determined using a **mass spectrometer**. The mass spectrum shows m/z values (mass-to-charge ratio) on the x-axis and relative abundance (or percentage abundance) on the y-axis. For elements, the spectrum reveals the isotopic composition. For molecules, the **molecular ion peak (M^+)** gives the relative molecular mass, and fragmentation patterns provide structural information.

The four stages of mass spectrometry are: (1) **Ionisation** — electrons are fired at the sample to knock off an electron, forming positive ions; (2) **Acceleration** — ions are accelerated by an electric field; (3) **Deflection** — ions are deflected by a magnetic field (lighter ions and higher-charged ions deflect more); (4) **Detection** — ions reaching the detector produce a signal proportional to abundance.

Worked Example — Calculating A_r from a Mass Spectrum:

A mass spectrum of boron shows two peaks: m/z = 10 (abundance 20%) and m/z = 11 (abundance 80%).

A_r = (10 × 20 + 11 × 80) / 100 = (200 + 880) / 100 = **10.8**

This explains why boron's A_r on the periodic table is 10.8, not a whole number.

Ions and Ionic Compounds

Ions form when atoms gain or lose electrons to achieve a stable electron configuration (usually a full outer shell, like a noble gas). Metals lose electrons to form positive cations. Non-metals gain electrons to form negative anions.

$$Na \rightarrow Na^+ + e^- \quad Cl + e^- \rightarrow Cl^-$$

Ionic compounds are held together by strong electrostatic attractions between oppositely charged ions arranged in a regular three-dimensional **giant ionic lattice**. Each ion is surrounded by ions of opposite

charge, maximising electrostatic attraction and minimising repulsion.

2.2 Amount of Substance

Mole: The amount of substance that contains exactly 6.022×10^{23} particles (atoms, molecules, ions, electrons, or formula units). This number is the **Avogadro constant, N_A**. One mole of any substance contains the same number of specified particles.

Molar Mass

$$n = m / M$$

Where n = amount in moles, m = mass in grams, M = molar mass in g mol^{-1}. The molar mass is numerically equal to the relative formula mass (M_r). For example, M_r of H_2O = (2 × 1.0) + 16.0 = 18.0, so its molar mass is 18.0 g mol^{-1}.

Empirical and Molecular Formulae

Empirical formula: The simplest whole-number ratio of atoms of each element in a compound. For example, the empirical formula of glucose ($C_6H_{12}O_6$) is CH_2O.

Molecular formula: The actual number of atoms of each element in one molecule of a compound. It is a whole-number multiple of the empirical formula.

To find the empirical formula from percentage composition or mass data:

1. Divide each mass (or percentage) by the element's A_r to get moles.
2. Divide all mole values by the smallest to get the simplest ratio.
3. If the ratio is not whole numbers (e.g. 1:1.5), multiply all values by the smallest integer needed (e.g. ×2 to get 2:3).

To find the molecular formula, divide the M_r by the empirical formula mass to find the multiplier n, then multiply each subscript in the empirical formula by n.

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Worked Example — Finding Empirical and Molecular Formula:

A compound contains 40.0% C, 6.7% H, and 53.3% O by mass. Its M_r = 180.0.

Step 1: Moles: C = 40.0/12.0 = 3.33, H = 6.7/1.0 = 6.7, O = 53.3/16.0 = 3.33
Step 2: Divide by smallest (3.33): C = 1, H = 2, O = 1
Step 3: Empirical formula = **CH_2O** (empirical formula mass = 30.0)
Step 4: n = 180.0 / 30.0 = 6
Step 5: Molecular formula = **$C_6H_{12}O_6$** (glucose)

Exam Tip: If the ratio after dividing by the smallest is not a whole number (e.g. 1:1.5 or 1:1.33), multiply all values by the smallest integer needed. 1:1.5 → ×2 → 2:3. 1:1.33 → ×3 → 3:4. Always show your working clearly.

Moles in Solutions

$$n = c \times V$$

Where c = concentration in mol dm^{-3} and V = volume in dm^3. Remember: to convert cm^3 to dm^3, divide by 1000. This equation is fundamental and appears in almost every quantitative chemistry question.

Moles in Gases

At room temperature and pressure (RTP: approximately 20 °C and 1 atm), one mole of any gas occupies approximately **24.0 dm^3** (24,000 cm^3). This is the molar gas volume.

$$n = V / 24.0 \quad (\text{at RTP, V in } dm^3)$$

The Ideal Gas Equation

$$pV = nRT$$

Where p = pressure in Pa, V = volume in m^3, n = moles, R = 8.314 J K^{-1} mol^{-1}, T = temperature in K.

Unit conversions:

- $cm^3 \rightarrow m^3$: divide by 10^6
- $dm^3 \rightarrow m^3$: divide by 1000
- °C → K: add 273 (or 273.15 for precision)
- kPa → Pa: multiply by 1000
- atm → Pa: multiply by 101325

Exam Tip: The most common error with pV = nRT is forgetting to convert units. **Always check**: pressure in Pa, volume in m^3, temperature in K. Write out the conversion explicitly to avoid losing marks. Also rearrange the equation before substituting values.

Worked Example — Ideal Gas Equation:

Calculate the volume occupied by 0.250 mol of CO_2 at 50 °C and 100 kPa.

T = 50 + 273 = 323 K; p = 100 × 1000 = 100,000 Pa
V = nRT / p = (0.250 × 8.314 × 323) / 100,000
V = 671.35 / 100,000 = 6.71×10^{-3} m^3 = **6.71 dm^3**

Concentration in g dm^{-3}

$$\text{concentration (g dm}^{-3}) = \text{concentration (mol dm}^{-3}) \times M_r$$

2.3 Acid-Base and Redox Reactions

Acids and Bases

Acid (Bronsted-Lowry): A proton (H^+) donor. In aqueous solution, acids release H^+ ions, which combine with water to form H_3O^+ (hydronium ions).

Base: A proton (H^+) acceptor. A soluble base is called an **alkali** and releases OH^- ions in aqueous solution. Common alkalis include NaOH, KOH, and $Ca(OH)_2$.

Strong acids (HCl, H_2SO_4, HNO_3) fully dissociate in water. **Weak acids** (CH_3COOH, H_2CO_3) only partially dissociate, establishing an equilibrium.

Key Acid Reactions

Reaction Type	General Equation	Example
Acid + Metal	Acid + Metal → Salt + Hydrogen	$2HCl(aq) + Mg(s) \rightarrow MgCl_2(aq) + H_2(g)$
Acid + Metal Oxide	Acid + Metal Oxide → Salt + Water	$2HCl(aq) + CuO(s) \rightarrow CuCl_2(aq) + H_2O(l)$
Acid + Metal Hydroxide	Acid + Metal Hydroxide → Salt + Water	$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
Acid + Carbonate	Acid + Carbonate → Salt + Water + CO_2	$2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

The **salt** formed depends on the acid used: HCl produces chlorides, H_2SO_4 produces sulfates, and HNO_3 produces nitrates.

Oxidation States (Oxidation Numbers)

Oxidation state: A number assigned to an atom to indicate the degree of oxidation or reduction it has undergone. It represents the number of electrons that an atom has effectively lost, gained, or shared unequally in a compound.

Rules for assigning oxidation states (in order of priority):

1. Uncombined (free) elements: **0** (e.g. Na, O_2, P_4, S_8)
2. Simple monatomic ions: equal to the charge ($Na^+ = +1$, $Cl^- = -1$, $Al^{3+} = +3$)
3. Fluorine: **always −1** (most electronegative element)

4. Oxygen: usually **−2** (except in peroxides [−1, e.g. H_2O_2] and in OF_2 [+2])
5. Hydrogen: usually **+1** (except in metal hydrides [−1, e.g. NaH])
6. The sum of oxidation states in a neutral compound = **0**
7. The sum of oxidation states in a polyatomic ion = **charge of the ion**

Worked Example — Finding Oxidation States:

Find the oxidation state of Mn in $KMnO_4$:

K = +1 (Group 1 metal), O = −2 each, total must = 0
(+1) + Mn + 4(−2) = 0
+1 + Mn − 8 = 0
Mn = **+7**

Redox Reactions

Oxidation: Loss of electrons / increase in oxidation state.
Reduction: Gain of electrons / decrease in oxidation state.
Remember: **OIL RIG** — Oxidation Is Loss, Reduction Is Gain (of electrons).

A redox reaction involves *simultaneous* oxidation and reduction. The substance that is oxidised is the **reducing agent** (electron donor); the substance that is reduced is the **oxidising agent** (electron acceptor).

Writing Redox Half-Equations and Ionic Equations

To write a balanced redox equation from half-equations:

1. Write separate half-equations for oxidation and reduction.
2. Balance atoms other than O and H first.
3. Balance O by adding H_2O to the side deficient in oxygen.
4. Balance H by adding H^+ to the side deficient in hydrogen.
5. Balance charge by adding electrons (e^-) to the more positive side.
6. Multiply half-equations so the electrons cancel when combined.

7. Add the half-equations together and simplify.

Worked Example — Balancing a Redox Equation:

Write a balanced equation for the reaction of Fe^{2+} with acidified MnO_4^-.

Reduction: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (Mn goes from +7 to +2)

Oxidation: $Fe^{2+} \rightarrow Fe^{3+} + e^-$ (Fe goes from +2 to +3)

Multiply the oxidation half-equation by 5 so electrons cancel:
$5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$

Overall: $MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$
Check: atoms balanced, charge balanced (both sides = +17).

Exam Tip: Always check that your final equation is balanced for both atoms AND charge. State symbols should be included where possible. A common error is forgetting to multiply one of the half-equations before combining.

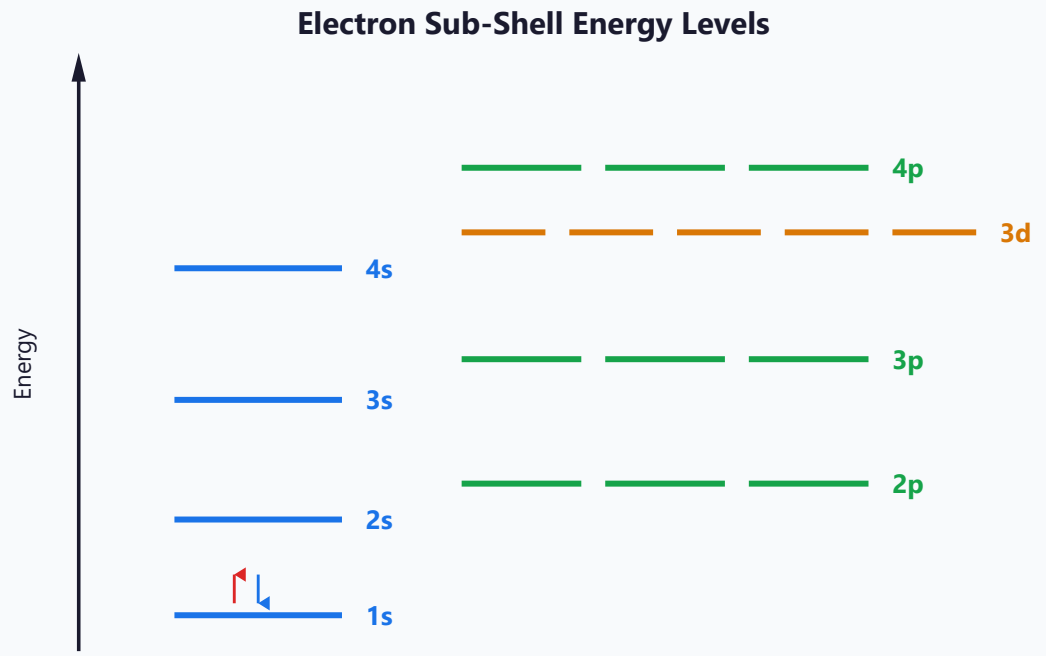
2.4 Electrons and Bonding

Electron Configuration

Electrons occupy orbitals in order of increasing energy (the **Aufbau principle**). Each orbital holds a maximum of 2 electrons with opposite spins (**Pauli exclusion principle**). Within a sub-shell, electrons occupy orbitals singly before pairing (**Hund's rule**). The order of filling is: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p.

Sub-shell	Number of Orbitals	Max Electrons	Shape
s	1	2	Spherical
p	3	6	Dumbbell (figure-of-eight)

d	5	10	Complex (mostly clover-leaf)



Filling order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p (4s fills before 3d but is lost first in ions)

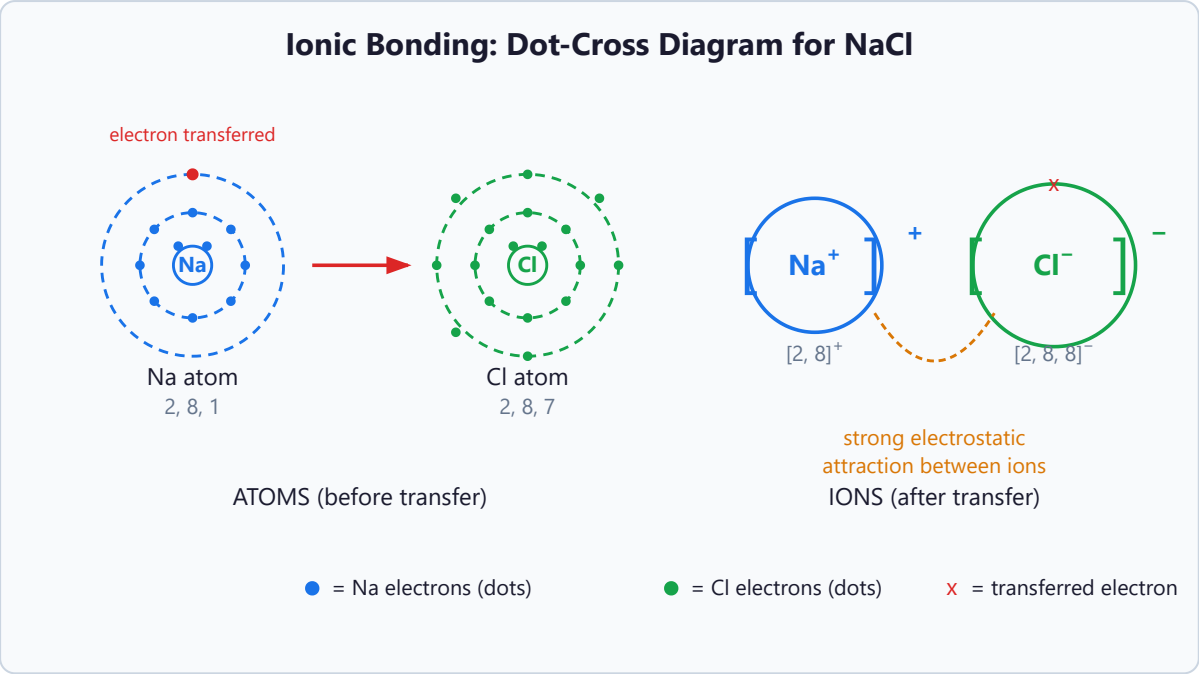
For d-block elements, the 4s sub-shell fills before 3d but is **lost first** when forming ions. For example, Fe is [Ar] $3d^6\ 4s^2$, but Fe^{2+} is [Ar] $3d^6$ (the two 4s electrons are removed first). Fe^{3+} is [Ar] $3d^5$.

Exceptions: Chromium is [Ar] $3d^5\ 4s^1$ (not $3d^4\ 4s^2$) and copper is [Ar] $3d^{10}\ 4s^1$ (not $3d^9\ 4s^2$). This is because half-filled ($3d^5$) and fully-filled ($3d^{10}$) d sub-shells have extra stability due to symmetrical electron distribution and favourable exchange energy.

Exam Tip: When writing the electron configuration of d-block ions, **always remove the 4s electrons before removing any 3d electrons**. This is the opposite of the filling order and is a very common exam question. Cr and Cu exceptions must be memorised.

Ionic Bonding

Ionic bond: The strong electrostatic attraction between oppositely charged ions in an ionic lattice, formed by the transfer of one or more electrons from a metal atom to a non-metal atom.



Properties of ionic compounds:

- **High melting and boiling points** — strong electrostatic forces between ions require a lot of energy to overcome
- **Conduct electricity when molten or dissolved** — ions are free to move and carry charge
- **Do not conduct when solid** — ions are fixed in position in the lattice and cannot move
- **Often soluble in polar solvents** like water — the polar water molecules can surround and stabilise the ions (hydration)
- **Brittle** — applying force shifts layers so like-charged ions align, causing repulsion and the crystal shatters

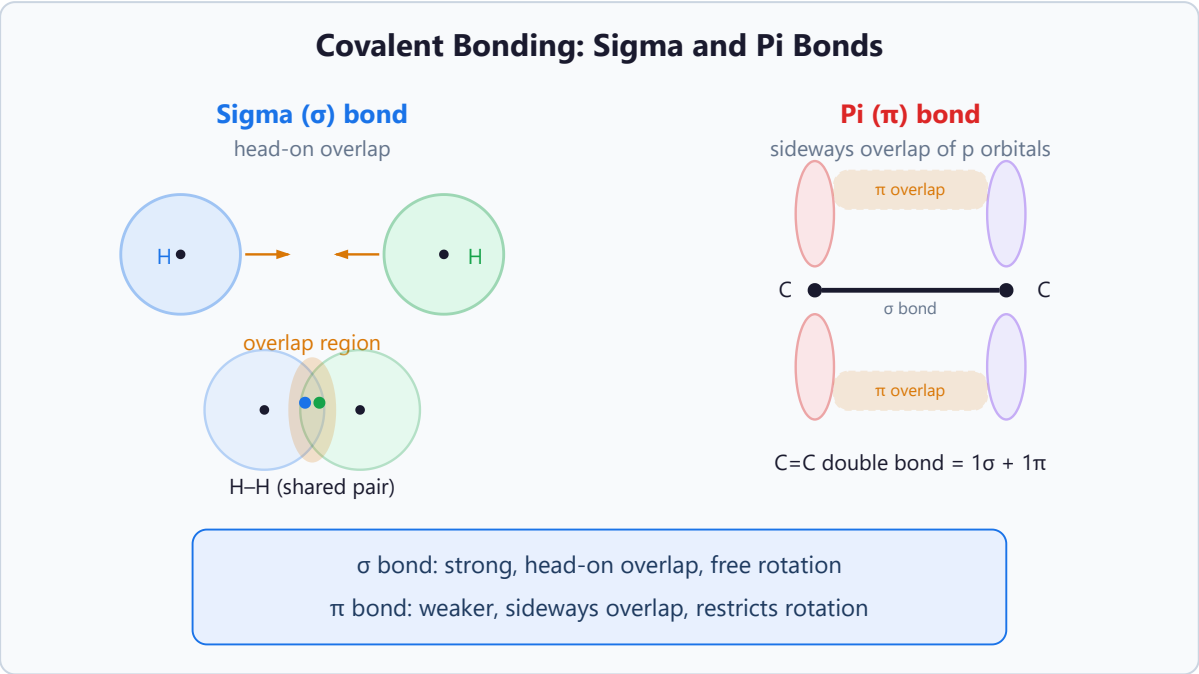
Covalent Bonding

> **Covalent bond:** A shared pair of electrons between two non-metal atoms, where both nuclei are attracted to the shared electron pair. This mutual attraction holds the atoms together. Each atom contributes one electron to the shared pair (in a normal covalent bond).

A **sigma (σ) bond** is formed by head-on overlap of orbitals along the internuclear axis. It is the strongest type of covalent bond and allows free rotation. A **pi (π) bond** is formed by sideways overlap of p orbitals above and below the internuclear axis. It is weaker than a sigma bond and restricts rotation.

- Single bond = 1 σ bond

- Double bond = 1 σ + 1 π bond
- Triple bond = 1 σ + 2 π bonds



A **dative (coordinate) bond** is a covalent bond where both electrons in the shared pair come from the same atom (the donor). It is shown as an arrow from the donor atom to the acceptor. Once formed, a dative bond is identical in strength to a regular covalent bond. Examples: NH_4^+ (nitrogen donates a lone pair to H^+), Al_2Cl_6 (Cl donates to Al), and $[Cu(NH_3)_4]^{2+}$ (NH_3 donates to Cu^{2+}).

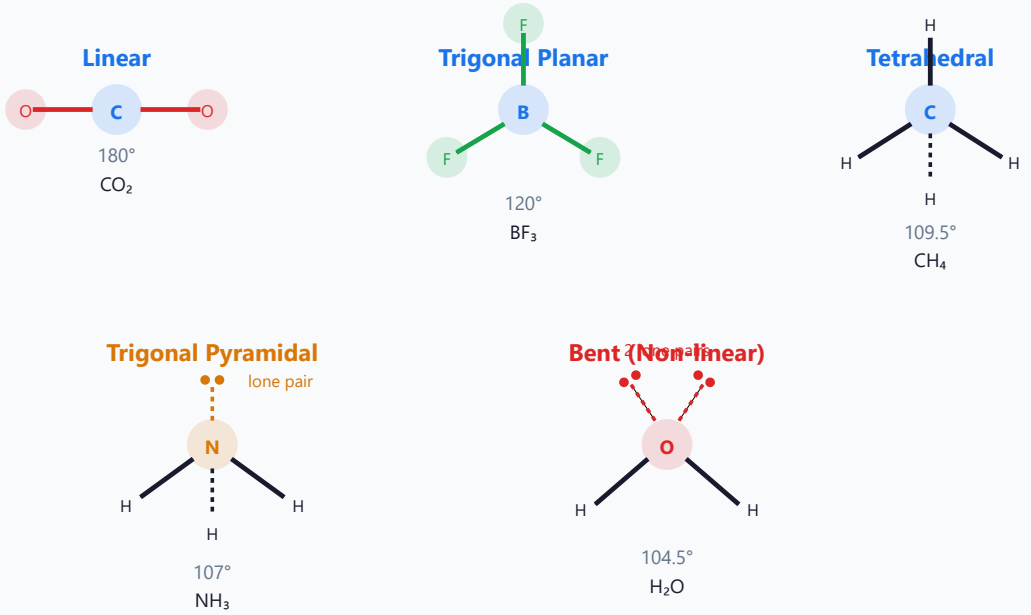
Shapes of Molecules — VSEPR Theory

Valence Shell Electron Pair Repulsion (VSEPR) theory predicts molecular shapes based on the principle that electron pairs around a central atom repel each other and arrange themselves as far apart as possible to minimise repulsion. The order of repulsion strength is:

Lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair

This means lone pairs compress bond angles below the ideal values.

Shapes of Molecules (VSEPR Theory)



Lone pairs repel more than bonding pairs, reducing bond angles: 109.5° → 107° → 104.5°

Electron Pairs	Bonding Pairs	Lone Pairs	Shape	Bond Angle	Example
2	2	0	Linear	180°	CO_2, $BeCl_2$, HCN
3	3	0	Trigonal planar	120°	BF_3, $AlCl_3$, NO_3^-
3	2	1	Bent (non-linear)	~117–120°	SO_2, NO_2^-
4	4	0	Tetrahedral	109.5°	CH_4, NH_4^+, CCl_4
4	3	1	Trigonal pyramidal	107°	NH_3, PCl_3, H_3O^+
4	2	2	Bent (non-linear)	104.5°	H_2O, H_2S
5	5	0	Trigonal bipyramidal	90° and 120°	PCl_5
6	6	0	Octahedral	90°	SF_6

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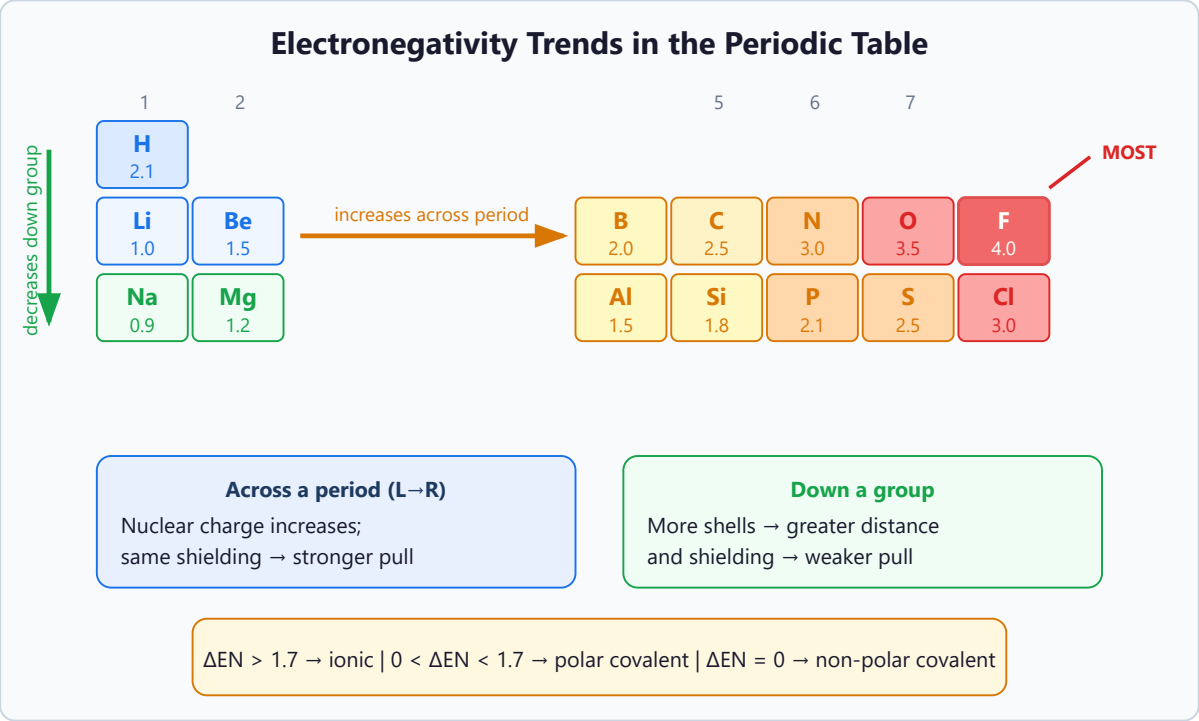
Exam Tip: To determine the shape: (1) draw the dot-cross diagram, (2) count the total electron pairs around the central atom, (3) identify how many are bonding and how many are lone pairs, (4) name the shape based on the arrangement of *atoms* (not electron pairs). Remember that double bonds count as one bonding region for VSEPR purposes.

Electronegativity and Bond Polarity

Electronegativity: The ability of an atom to attract the bonding pair of electrons in a covalent bond towards itself. Measured on the Pauling scale, where fluorine is the most electronegative element (4.0). Electronegativity increases across a period (greater nuclear charge, same shielding) and decreases down a group (greater distance and shielding).



If there is a difference in electronegativity between bonded atoms, the bond is **polar** — the more electronegative atom carries a partial negative charge (δ−) and the less electronegative atom carries a partial positive charge (δ+). A large electronegativity difference (>1.7) typically results in ionic bonding. A small difference gives a polar covalent bond.

If the molecule is **symmetrical** (e.g. CO_2, CCl_4, BF_3), the individual bond dipoles cancel out and the molecule has no overall dipole — it is non-polar overall, even if individual bonds are polar. If the molecule

is **asymmetrical** (e.g. $CHCl_3$, H_2O), the dipoles do not cancel and the molecule is polar.

Metallic Bonding

Metallic bond: The strong electrostatic attraction between positive metal ions (cations arranged in a regular lattice) and a "sea" of delocalised electrons that have been released from the outer shells of the metal atoms.

Properties of metals explained by metallic bonding:

- **High melting points** — strong electrostatic forces require a lot of energy to overcome. Stronger with more delocalised electrons and smaller ionic radius (e.g. Mg has a higher melting point than Na).
- **Good electrical conductivity** — delocalised electrons are free to move through the structure, carrying charge.
- **Good thermal conductivity** — delocalised electrons transfer kinetic energy rapidly through the structure.
- **Malleable and ductile** — layers of ions can slide over each other without breaking bonds, because the delocalised electron sea can adjust to the new positions.
- **Insoluble in water** — metallic bonds are too strong to be disrupted by water molecules.

Intermolecular Forces

Intermolecular forces act *between* molecules and are much weaker than covalent bonds *within* molecules. They determine physical properties such as boiling point, solubility, and viscosity. There are three types, in increasing order of strength:

Type	Strength	Occurs Between	Key Details
London (dispersion) forces	Weakest (but increases with M_r)	ALL molecules (including non-polar)	Caused by temporary fluctuations in electron density creating instantaneous dipoles that induce dipoles in neighbouring molecules. Strength increases with number of electrons (larger M_r = more electrons = stronger London forces). This explains why boiling points increase up a homologous series.

Permanent dipole–dipole forces	Moderate	Polar molecules	Attraction between the δ+ end of one polar molecule and the δ− end of another. Exist in addition to London forces. Examples: propanone, chloromethane.
Hydrogen bonding	Strongest intermolecular force	Molecules with N–H, O–H, or F–H bonds	A special, strong type of dipole interaction. Requires: (1) a hydrogen atom bonded to a highly electronegative atom (N, O, or F), and (2) a lone pair on a nearby N, O, or F atom. Explains the anomalously high boiling points of H_2O, NH_3, and HF compared to hydrides of the same group. Also explains ice being less dense than water (open lattice structure in ice), and the high surface tension and specific heat capacity of water.

Exam Tip: When explaining boiling point trends, always: (1) **name** the specific intermolecular forces involved in each substance, (2) **explain** how their strength differs and why, (3) **link** this to the energy required to overcome them. *Never* say "bonds are broken" when discussing boiling — it is intermolecular forces that are overcome, not covalent bonds. Use precise language: "more energy is required to overcome the stronger hydrogen bonds in water compared to the weaker London forces in methane."